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The Neuro-Evolutionary Mirror Paradox: Subcortical Motion Bias, the Self-Face Advantage, and the Dissociative Architecture of the Dialogical Brain

The interaction between the mammalian visual system and the reflective surface represents one of the most complex feedback loops in evolutionary biology. When a human or higher mammal encounters their own moving reflection, they are subjected to a multi-layered cognitive "trap" that engages primitive subcortical survival mechanisms, sophisticated cortical self-processing networks, and deep-seated internal dialogical structures. This phenomenon, which makes it nearly impossible to look away from a moving object in a mirror, is not merely a matter of curiosity but an ingrained neurological wiring that prioritizes the self as the most behaviorally salient object in the environment.¹

The scientific validity of this "magnetic" gaze is found in the dual-stream processing of the visual system, where the phylogenetically older midbrain structures react to motion and salience before the newer mammalian cortex can interpret the identity of the reflection.² This report examines the neurological, psychological, and physical dimensions of this mirror-lock phenomenon, structured through the lens of an eight-pane psychological matrix that details the progression from reflexive orienting to dissociative fragmentation.

The Tectofugal Pathway: Subcortical Primacy in Motion Detection

The initial inability to look away from a moving object in a mirror, such as during the peripheral flicker of a rearview mirror or the sudden movement in a bathroom vanity, is governed by the superior colliculus (SC), a layered midbrain structure.² In all vertebrates, visual signals from each field project to the opposite midbrain tectum, which in mammals is referred to as the superior colliculus.² This structure acts as the brain's "rapid detector" and "first responder," specialized for life-or-death situations where time is of the essence.⁴

The Mechanism of the Orienting Reflex

The superior colliculus maintains a topographical salience map of the environment, used to prioritize targets for action.² When movement occurs, the superficial layers of the SC receive

direct retinal input via the retinotectal pathway.² This pathway operates independently of the slower cortical visual system, providing a fast, reflexive orienting response to external (exogenous) stimuli.²

The SC does not merely detect motion; it computes the vector required for a saccade—a rapid eye movement—to align the fovea with the stimulus.⁵ In mammals like rodents, cats, and monkeys, visually guided motor actions are generated by these midbrain circuits.⁵ Recent research has identified "motion direction columns" within the SC, where neurons responding to the same orientation of motion cluster together.⁷ This system functions like a high-speed motion detection camera, reporting both location and direction to the motor system to facilitate rapid interception.⁷

Visual Pathway	Anatomical Route	Primary Function	Processing Speed
Subcortical (Retinotectal)	Retina → Superior Colliculus	Fast reflexive orienting; motion detection; survival response	Ultra-Rapid (First Responder) ²
Cortical (Geniculostriate)	Retina → LGN → Primary Visual Cortex	High visual acuity; color/form integration; object identification	Slower (Assessment/Adjustment) ³
Dorsal Stream	Visual Cortex → Parietal/Frontal Cortex	Spatial awareness; "Where" and "How" of objects	Integrated with collicular signals ²

The "Magnetic" Saccade

Because the mirror provides a perfectly synchronized moving target, the SC discharge is maximized. Saccade-related activity in the primate SC depends heavily on the presence of a visual target at the saccade endpoint.⁸ When a target is present, the presaccadic motor activity is at its most intense.⁸ In the case of a mirror, the moving stimulus (the self) is always present at the point of fixation, creating a continuous drive to maintain foveal alignment.⁸ This explains why mammals cannot look away: the SC is locked into a high-intensity discharge loop that prioritizes the moving reflection as the most salient environmental feature.³

The Neuropsychology of the Self-Face Advantage

Once the subcortical system has locked the gaze onto the mirror, the cortical "self-face advantage" takes over, reinforcing the fixation through high-level cognitive prioritization. The self-face advantage refers to the phenomenon where healthy individuals process images of their own faces faster and more accurately than those of others.¹

Subliminal Prioritization and the MedPFC

In healthy individuals, the self-face is a unique neural marker that is automatically detected in the early stages of visual processing.¹ Using Continuous Flash Suppression (CFS) to measure unconscious processing, researchers found that self-face images break through interocular inhibition and enter conscious awareness significantly faster than the faces of famous people or strangers.¹

This prioritization is mediated by the Medial Prefrontal Cortex (MedPFC) and the Anterior Cingulate Cortex (ACC).¹ The MedPFC is heavily involved in the integration of self-perception and self-awareness, while the right ACC exhibits greater responses when an individual processes their own face compared to a stranger's.¹ This neural bias creates a psychological "hook"; the brain is hardwired to allocate its limited attentional resources to the "self," making the mirror reflection a cognitive vacuum that pulls in the observer's focus.¹

Featural Scrutiny vs. Holistic Processing

Interestingly, the brain processes the self-face differently than it processes other faces. While other-face recognition relies on holistic patterns, mental images of one's own face are generated more quickly through a featural or local processing bias.¹⁰ We are more synchronized with the retrieval of specific facial features (eyes, nose, blemishes) of our own face than the overall configural pattern.¹⁰ This local processing bias explains the tendency to "scan" the mirror reflection for minute changes—a behavior that prolongs the duration of the gaze as the brain meticulously analyzes individual components of the face.¹⁰

Brain Region	Functional Role in Mirror Scrutiny	Psychological Impact
MedPFC	Self-processing and integration ¹	Foundation of self-awareness and self-advantage
ACC	Response to self-face vs. others ¹	Emotional salience and prioritization of the

		reflection
Right DLPFC	Prefrontal attentional systems ¹	High-level alertness; potential for detachment if over-activated
Right Posterior Insula	Self-attribution/egocentric reference ¹	Sense of body ownership and "being" the reflection

The Dialogical Self: The Internal Monologue as a Mirror Interlocutor

A critical reason why individuals find it difficult to look away from their reflection is the activation of the internal monologue. The mirror provides a physical partner for the "polyphonic self"—a multiplicity of inner voices representing different "I-positions" or points of view.¹²

Talking to the Person in the Brain

According to Dialogical Self Theory, intrapersonal communication (self-talk) involves an exchange of thoughts between different parts of the self.¹² When we look in a mirror, the "I" (the observer) interacts with the "Me" (the objectified reflection).¹² This interaction serves several vital functions:

1. **Self-Management and Regulation:** We use self-talk to "coach" ourselves ("Try to stay calm," "You look ready").¹²
2. **Social Assessment:** We use the reflection to simulate how others might perceive us, essentially "rehearsing" social interactions.¹²
3. **Self-Criticism:** The mirror often acts as a catalyst for a critical inner voice, especially in an age of digital self-objectification, which can lead to mirror-fixation driven by anxiety.¹¹

This inner speech is not a metaphor; it is the experience of silent, verbal thinking that utilizes the brain's language and theory-of-mind (ToM) centers.¹⁴ Because the mirror provides a perfectly synchronized "other" that responds to every motor command, the brain is tricked into a social loop. We talk to the person in the mirror because our neural architecture for communication is being stimulated by a realistic, responsive interlocutor.¹²

The Anxiety of the "Other"

For many, this internal dialogue is a constant presence that narratizes daily experience and reinforces beliefs.¹⁷ However, when the internal monologue is overly critical or distorted by anxiety, the mirror becomes a source of stress.¹⁶ A negative self-opinion acts as a "clouded

lens," casting a shadow on self-confidence and making the reflection appear as an adversary rather than an ally.¹⁶ This can result in a compulsive need to scrutinize the mirror in an attempt to "fix" or "manage" the perceived negative traits of the "other" in the brain.¹¹

The Strange-Face-in-the-Mirror Illusion and Sensory Deprivation

If the gaze remains locked for several minutes—particularly in dimly lit environments—the psychological effects transition from simple self-reflection into profound dissociative and hallucinatory states. This is known as the "Strange-Face-in-the-Mirror" illusion.¹³

The Mechanism of Troxler's Fading

The fundamental cause of these illusions is Troxler’s fading, an optical phenomenon where unchanging stimuli in the peripheral vision fade and disappear.²⁰ Under normal circumstances, involuntary fixational eye movements (microsaccades and drifts) prevent the adaptation of neurons to an unchanging environment.²¹ However, during intense, prolonged fixation on the eyes of a mirror reflection, these microsaccades may fail to refresh the retinal image sufficiently.²¹

As the neurons adapt, parts of the reflected face begin to vanish, leaving behind a distorted or blurred image.²⁰ The brain, which is hardwired to "look for faces" in any pattern, begins to over-interpret these blurred features.¹⁸

Apparitions and Dissociation

In experiments conducted by Giovanni Caputo, 50 healthy adults stared into a mirror for ten minutes in a dimly lit room.¹⁸ The results showed that 100% of participants experienced strange-face apparitions, often within the first minute.¹³

Type of Apparition	Reported Frequency	Psychological Response
Huge Deformations of Own Face	66%	Inquietude; sense of things out of control ¹⁸
Monstrous/Fantastical Faces	48%	Fear, disturbance, or astonishment ¹⁸
Unknown Person/Strange	28%	Sensation of otherness; enigmatic expression ¹⁸

'Other'		
Archetypal/Deceased Relatives	18% - 28%	Silent query; emotional connection or unease ¹⁸
Animal Faces (Cats, Lions)	18%	Astonishment; confusion of identity ¹⁸

These apparitions are symptoms of dissociation—a departure from the subject's connection to reality.¹⁹ Participants often felt that the "other" in the mirror was watching them with an independent agency, leading to intense emotional reactions.¹⁸ This effect is even more dramatic in patients with schizophrenia or schizotypal traits, who perceive more intense and frequent apparitions.²³

Archeo-Futurism: Denisovan-Neanderthal Symbolism and Dieselpunk Design

The user's request for a "Denisovan Neanderthal futuristic" style in the 1920s aesthetic (Dieselpunk) provides a rich visual metaphor for the evolution of human self-awareness. This "retro-futurism" blends the grimy, fuel-powered machinery of the early 20th century with the burgeoning symbolic mind of our ancient ancestors.²⁵

The Denisovan Chlorite Bracelet: A Paradigm of Ancient Tech

Central to this aesthetic is the 40,000-year-old Denisovan bracelet found in the Altai Mountains.²⁸ This artifact, made of polished green chlorite, demonstrates technology far ahead of its time, including high-speed easel drilling and precise grinding.²⁸ The bracelet's optical properties are particularly mirror-like: it reflects sunlight with a brilliant sparkle but casts a deep green hue in firelight.²⁹ This duality—between the clear reflection and the dark, distorted shadow—parallels the mirror-gazing experience.

Neanderthal Symbolism and Symbolic Thinking

Contrary to the myth of the "knuckle-dragging brute," Neanderthals were capable of symbolic thinking and ritual.³¹ They created the world's oldest cave art, including hand stencils, geometric designs, and painted stalactites, more than 64,000 years ago.³¹ The use of red ochre and black pigments in these "meaningful places" shows a mind starting to externalize internal cognitive processes into symbolic representations.³¹

In a Dieselpunk context, this prehistoric symbolism is integrated into 1920s industrialism. Imagine the "Mezhirich" style mammoth-bone huts—structures built from the skeletons of at least 95 mammoths—reimagined as the foundations for Art Deco skyscrapers.³³ These

structures, which sometimes featured "drums" made of mammoth skulls painted with ochre dots, represent the earliest examples of "monumental architecture" and social complexity.³³

1920s Dieselpunk: The "Decodence" Aesthetic

The 1920s futurist vision, often called "Ottensian Dieselpunk," rejoices in the enthusiasm of the Jazz Age while acknowledging the grimy reality of oil and steel.²⁶ Key visual motifs include:

- **Riveted Surfaces:** Large, functional components like the "Mezhirich" bone structures held together by steel rivets and gears.²⁵
- **Aerodynamic Shapes:** Rectilinear lines married to the "Streamline Moderne" curves of early aircraft and zeppelins.²⁶
- **Noir Lighting:** The gritty, rain-slicked atmosphere of film noir, where neon lights reflect off dark, oily puddles.²⁵

The Eight-Pane Psychological Matrix of Mirror Interaction

To depict the scientific and psychological progression of mirror-gazing, we divide the experience into eight "window panes." Each pane represents a distinct phase of the mammalian brain's response to its reflection.

Pane 1: The Subcortical Trigger (Orienting Reflex)

The first pane depicts the "Mammalian First Responder." As the subject passes a mirror, the **superior colliculus** detects the peripheral motion.² The visual field shows a sudden spike in salience. The brain, acting on a primitive circuit, triggers a reflexive saccade. The subject cannot look away because the midbrain has already committed to foveal alignment with the "threat/opportunity" in the mirror.⁶

Pane 2: The Self-Face Lock (MedPFC Activation)

The second pane shows the "Neural Mirror." The **Medial Prefrontal Cortex** and **Anterior Cingulate Cortex** recognize the image as the self.¹ The "Self-Face Advantage" activates, creating a massive attentional surge.¹ The brain enters a supraliminal state where the reflection is prioritized over the entire room. The gaze becomes "locked" by the neural reward of self-recognition.¹

Pane 3: The Dialogical Emergence (Internal Monologue)

The third pane depicts "The Person in the Brain." The **Dialogical Self** engages, and the internal monologue begins to talk to the reflection.¹² The subject sees their own lips in the mirror and hears a subvocalized voice narrating their thoughts.¹⁵ The reflection is no longer an image but an interlocutor. The anxiety of "looking right" or "saying the right thing" keeps the

subject pinned to the glass.¹²

Pane 4: The Onset of Fading (Troxler Effect)

The fourth pane shows "The Fading World." As fixation continues, **Troxler's Fading** begins.²⁰ The subject's peripheral features start to blur and dissolve into the background.²¹ The neural signal for unchanging stimuli weakens. The room around the subject disappears, leaving only the eyes and the core features of the face as a solitary island in a gray void.²⁰

Pane 5: The Pareidolia Threshold (Face Detection Overload)

The fifth pane depicts "The Search for Meaning." Sensing the loss of sensory data, the **face detection system** over-interprets the blurred reflection.¹⁸ The brain begins to "fill in" the missing features with archetypal data.¹⁸ Shadowed areas on the cheeks become hollows; the skin takes on the texture of ancient stone or "Denisovan" green chlorite.³⁰

Pane 6: The Strange-Face Apparition (Hallucinatory State)

The sixth pane shows "The Monster in the Mirror." The subject experiences a full **strange-face apparition**.¹⁸ Their own face is replaced by a monstrous or animalistic entity.¹³ The "other" appears to have its own agency, smiling or grinding its teeth. The subject feels intense "inquietude" as the biological boundary between "Self" and "Other" collapses.¹⁸

Pane 7: The Dissociative Break (Depersonalization)

The seventh pane depicts "The Fragmented Self." The subject enters a state of **depersonalization/derealization (DPD)**.¹ The reflection is no longer "Me"; it is a stranger watching from a distant plane.¹⁸ The limb ownership structures (posterior insula) are inhibited.¹ The subject feels like a bystander to their own reflection, an "extinct hominin" trapped in a futuristic 1920s machine.¹

Pane 8: The Recursive Loop (Infinite Feedback)

The eighth pane shows "The Infinite Sink." The subject is trapped in a **recursive loop** of self-observation and self-dialogue.¹² The physics of the mirror—represented by light bouncing infinitely in the negative space—mirrors the psychological trap.⁴⁰ The brain cannot look away because the "Self" has become an infinite regress of questions and observations, echoing back to its origin.¹³

Optical Physics: The Infinite Bouncing Light in Negative Space

The physical foundation of this visual trap is the behavior of photons between reflective

surfaces. According to the **Law of Reflection**, light waves strike the mirror and reflect at an equal angle.⁴¹

$$\theta_{\text{incident}} = \theta_{\text{reflected}}$$
When multiple mirror planes are present, as in the requested image, light enters a state of "infinite reflection".⁴⁰ In the "negative space" between the window panes, the photon acts like a pool ball, bouncing and refracting without ever repeating its path exactly.⁴⁰

The Path of the Photon

The index of refraction ($n = \frac{c}{v}$) in the glass causes a slight bend, but the primary phenomenon is the specular reflection from the metallic backing.⁴¹ In an "infinity mirror" setup, each bounce adds a new virtual image, but each reflection also loses a fraction of light energy due to absorption.⁴⁰

Physical Property	Value/Effect	Visual Result
Index of Refraction (n)	~ 1.5 for glass ⁴⁴	Slight displacement of the virtual image
Reflectivity	$>90\%$ for silvered mirrors ⁴¹	Multiple visible reflections
Light Absorption	Cumulative with each bounce ⁴⁰	Distant reflections appear dimmer and greener
Virtual Image Depth	$d_i = d_o$ ⁴⁴	Sense of depth behind the mirror surface

In the "negative space" between the panes, a single streak of light followed through all the space back to its origin represents the **Huygens-Fresnel principle**, where all secondary waves add up to a coherent path.⁴² The light bounces and refracts just enough to create a complex web of energy that never overlaps, illustrating the brain's own non-linear processing of visual data.⁴²

Conclusion: The Evolutionary Necessity of the Gaze

The scientific evidence confirms that the human inability to look away from a mirror is a result of a highly efficient, subcortical-cortical "pincer maneuver".¹ The superior colliculus locks the eyes onto motion, while the MedPFC and the Dialogical Self lock the mind onto identity.⁷ This

"wiring" likely evolved as an advantage for social survival, allowing early hominins like Neanderthals and Denisovans to monitor their own social signaling and maintain identity within complex groups.²⁷

The mirror, therefore, is not a passive object. It is a biological catalyst that triggers a cascade of effects—from reflexive orienting to hallucinatory dissociation—revealing the fragile, multi-layered nature of the mammalian brain.¹⁸

Image Generator Prompt Instructions

To accurately depict this habit/ingrained wiring, use the following prompt structure derived from this research:

Style: Denisovan-Neanderthal Futuristic (1920s Dieselpunk / "Decodence").

Subject: Eight horizontal window panes showing the progression of a Neanderthal man (Jazz Age attire) looking into a mirror.

Pane 1-8: Follow the "Eight-Pane Psychological Matrix" descriptions above.

Negative Space: A single streak of brilliant white light (a photon) bouncing like a pool ball through all the negative space between the panes, following the Law of Reflection ($\theta_i = \theta_r$) and Snell's Law ($n_1 \sin \theta_1 = n_2 \sin \theta_2$), eventually returning to its origin without overlapping.

Materials: Mammoth bone structure frames, green chlorite "Denisovan" bracelet textures, riveted steel, vacuum tubes, and Art Deco pinstripes.

Atmosphere: Gritty industrial noir, heavy oil-stained machinery, glowing red ochre symbols on the frames, and infinite reflections receding into a dark green firelit void.

This configuration will show people exactly how the mammalian brain works when confronted with the "virtual other" in the mirror—a loop of survival, identity, and recursive thought from which it is impossible to look away.

Works cited

1. An Experimental Study of Subliminal Self-Face Processing in ... - NIH, accessed December 20, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC9775423/>
2. Seeing without a Scene: Neurological Observations on the Origin and Function of the Dorsal Visual Stream - PMC - PubMed Central, accessed December 20, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11121949/>
3. Seeing without a Scene: Neurological Observations on the Origin and Function of the Dorsal Visual Stream - ResearchGate, accessed December 20, 2025, https://www.researchgate.net/publication/380585476_Seeing_without_a_Scene_Neurological_Observations_on_the_Origin_and_Function_of_the_Dorsal_Visual_Stream
4. Fast Detector/First Responder: Interactions between the Superior Colliculus-Pulvinar Pathway and Stimuli Relevant to Primates - Frontiers, accessed December 20, 2025,

- <https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2017.00067/full>
5. Brain-wide arousal signals are segregated from movement planning in the superior colliculus - eLife, accessed December 20, 2025, <https://elifesciences.org/reviewed-preprints/99278>
 6. Superior colliculus drives stimulus-evoked directionally biased saccades and attempted head movements in head-fixed mice - PubMed Central, accessed December 20, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC8747496/>
 7. Motion detector in the brain translates visual information into movement, accessed December 20, 2025, <https://www2.mrc-lmb.cam.ac.uk/motion-detector-in-the-brain-translates-visual-information-into-movement/>
 8. Dependence of Saccade-Related Activity in the Primate Superior Colliculus on Visual Target Presence - American Physiological Society Journal, accessed December 20, 2025, <https://journals.physiology.org/doi/10.1152/jn.2001.86.2.676>
 9. Visual Functions of the Primate Superior Colliculus - Annual Reviews, accessed December 20, 2025, <https://www.annualreviews.org/content/journals/10.1146/annurev-vision-111022-123817?TRACK=RSS>
 10. Not All Faces Are Processed Equally: Evidence for Featural Rather Than Holistic Processing of One's Own Face in a Face-Imaging Task - ResearchGate, accessed December 20, 2025, https://www.researchgate.net/publication/24187240_Not_All_Faces_Are_Processed_Equally_Evidence_for_Featural_Rather_Than_Holistic_Processing_of_One's_Own_Face_in_a_Face-Imaging_Task
 11. Mirror Meditation: Tara Well, P D | PDF | Self Awareness | Anxiety - Scribd, accessed December 20, 2025, <https://www.scribd.com/document/636540721/Untitled>
 12. Types of Inner Dialogues and Functions of Self-Talk: Comparisons and Implications, accessed December 20, 2025, <https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2020.00227/full>
 13. Visual Perception during Mirror-Gazing at One's Own Face in Patients with Depression - NIH, accessed December 20, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4258311/>
 14. The brain's conversation with itself: neural substrates of dialogic inner speech - PMC, accessed December 20, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4692319/>
 15. Exploring the Inner Monologue: Conversations with Myself - Lemon8-app, accessed December 20, 2025, https://www.lemon8-app.com/@julove13_333/7449342292730577454?region=us
 16. The Mirror Effect: Unveiling the Connection Between Self-Opinion and Self-Confidence, accessed December 20, 2025, <https://theconfidencetrack.com/the-mirror-effect-unveiling-the-connection-between-self-opinion-and-self-confidence/>

17. Why Your Inner Monologue Isn't the Whole Story | Self-Awareness & Counselling, accessed December 20, 2025, <https://helpscounselling.com/blog/why-your-inner-monologue-isn-t-the-whole-story>
18. The strange-face-in-the-mirror illusion – Mind Hacks, accessed December 20, 2025, <https://mindhacks.com/2010/09/18/the-strange-face-in-the-mirror-illusion/>
19. Strange face illusions – University of Hertfordshire Research Archive, accessed December 20, 2025, https://uhra.herts.ac.uk/id/eprint/10250/1/1_s2.0_S105381002300017X_main.pdf
20. Troxler's fading – Wikipedia, accessed December 20, 2025, https://en.wikipedia.org/wiki/Troxler%27s_fading
21. THE ROLE OF FIXATIONAL EYE MOVEMENTS IN VISUAL PERCEPTION – Neuralcorrelate.com, accessed December 20, 2025, https://www.neuralcorrelate.com/smc/files/publications/martinez-conde_et_al_nrn04.pdf
22. Eye movements under various conditions of image fading | JOV – Journal of Vision, accessed December 20, 2025, <https://jov.arvojournals.org/article.aspx?articleid=2158180>
23. Strange-Face-in-the-Mirror Illusion and Schizotypy During Adolescence – PMC – NIH, accessed December 20, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4373638/>
24. Dazed And Confused: 10 Minutes Of Eye Contact Alters State Of Consciousness, Without Drugs – Medical Daily, accessed December 20, 2025, <https://www.medicaldaily.com/dazed-and-confused-10-minutes-eye-contact-alters-state-consciousness-without-drugs-349996>
25. What is Dieselpunk? (Plus 5 Examples) – Imagine Forest, accessed December 20, 2025, <https://www.imagineforest.com/blog/what-is-dieselpunk/>
26. Dieselpunk – Wikipedia, accessed December 20, 2025, <https://en.wikipedia.org/wiki/Dieselpunk>
27. A Tale of Two Species: The Origins of Art and the Neanderthal Challenge – ResearchGate, accessed December 20, 2025, https://www.researchgate.net/publication/329959150_A_Tale_of_Two_Species_The_Origins_of_Art_and_the_Neanderthal_Challenge
28. 40,000-Year-Old Bracelet Made by Extinct Human Species Discovered in Siberia, accessed December 20, 2025, <https://unbelievable-facts.com/2019/09/bracelet-made-by-extinct-human-species.html>
29. Stone Bracelet May Have Been Made by Denisovans – Archaeology Magazine, accessed December 20, 2025, <https://archaeology.org/news/2015/05/07/150507-siberia-denisovan-bracelet/>
30. Stone bracelet is oldest ever found in the world | Ancient Origins, accessed December 20, 2025, <https://www.ancient-origins.net/news-history-archaeology/stone-bracelet-oldest-ever-found-world-003027>
31. Study reveals earliest cave art belonged to Neanderthals, not humans – The

- Journal, accessed December 20, 2025,
<https://www.thejournal.ie/cave-art-neanderthals-3868191-Feb2018/>
32. Cross-Modality Information Transfer: A Hypothesis about the Relationship among Prehistoric Cave Paintings, Symbolic Thinking, and the Emergence of Language - Frontiers, accessed December 20, 2025,
<https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2018.00115/full>
 33. Perfectly Preserved Mammoth Bone Dwellings Found in Ukraine May Be the Earliest Examples of Architecture - Earthly Mission, accessed December 20, 2025,
<https://earthlymission.com/mezhyrich-perfectly-preserved-mammoth-bone-dwellings-shelters-ukraine-earliest-examples-architecture/>
 34. Mezhyrich / Межиріч - Mammoth Camp - Don's Maps, accessed December 20, 2025, <https://donsmaps.com/mammothcamp.html>
 35. Mezhyrich - Upper Paleolithic Mammoth Bone Settlement in Ukraine - ThoughtCo, accessed December 20, 2025,
<https://www.thoughtco.com/mezhyrich-mammoth-bone-settlement-171805>
 36. Dieselpunk | Aesthetics Wiki - Fandom, accessed December 20, 2025,
<https://aesthetics.fandom.com/wiki/Dieselpunk>
 37. Steampunk vs. Dieselpunk: A Complete Comparison - VSD - Verdine Daniels, accessed December 20, 2025,
<https://verlinedaniels.com/blogs/news/steampunk-vs-dieselpunk-a-complete-comparison-key-differences-aesthetics>
 38. (PDF) Preliminary Evidence of a Missing Self Bias in Face Perception FOR Individuals with Dissociative Identity Disorder - ResearchGate, accessed December 20, 2025,
https://www.researchgate.net/publication/329028170_Preliminary_Evidence_of_a_Missing_Self_Bias_in_Face_Perception_FOR_Individuals_with_Dissociative_Identity_Disorder
 39. Perceptual Illusions - Understanding and Overcoming Hate, accessed December 20, 2025, <https://www.overcominghateportal.org/perceptual-illusions.html>
 40. Infinity mirror - Wikipedia, accessed December 20, 2025,
https://en.wikipedia.org/wiki/Infinity_mirror
 41. Introduction to the Reflection of Light - Evident Scientific, accessed December 20, 2025,
<https://evidentscientific.com/en/microscope-resource/knowledge-hub/lightandcolor/reflectionintro>
 42. Reflection (physics) - Wikipedia, accessed December 20, 2025,
[https://en.wikipedia.org/wiki/Reflection_\(physics\)](https://en.wikipedia.org/wiki/Reflection_(physics))
 43. An Infinite Number of Reflections? Really? | STEM Activity - Science Buddies, accessed December 20, 2025,
<https://www.sciencebuddies.org/stem-activities/an-infinite-number-of-reflections-really>
 44. Physics 2233 : Chapter 32 Examples : Light Reflection and Refraction NOTE, accessed December 20, 2025,
<https://newton.ph.msstate.edu/fox/ph2233/examples32.pdf>